

CC Lab 14

Task 1:

```
t1 = x + y
if t1 < 10 goto L1
t2 = t1 * 2
goto L2
L1: t2 = t1 - 5
L2: t2z = t2 + 1
```

Step-by-Step Stack Code:

```
PUSH x
PUSH y
ADD
POP t1
```

Condition check: if $t1 < 10$ goto L1

```
PUSH t1
PUSH 10
JIMPLT L1
```

if condition is FALSE:

```
PUSH t1
PUSH 2
MUL
POP t2
```

GIVENCHY

Label L1 (if condition TRUE):

LABEL L1

PUSH t1

PUSH 5

SUB

POP t2

Label L2:

LABEL L2

PUSH t2

PUSH 1

ADD

POP z

Task 2:

$$a = b + c$$

$$d = a - e$$

$$f = d * 2$$

Generated Code:

Step 1: $a = b + c$

LOAD R0, b

LOAD R1, c

ADD R0, R1

STORE a, R0

Step 2: $d = a - e$

LOAD R_0, a

LOAD R_1, e

SUB R_0, R_1

STORE d, R_0

Step 3: $f = d * 2$

LOAD R_0, d

MOVE $R_1, 2$

MUL R_0, R_1

STORE f, R_0